Calculus II

Solution to Assignment 5

Part I. Regular Problems

1. Which of the following vectors are unit vectors?
   (a) $\frac{1}{50}(3,4,5)$: **No**, since $\left\|\frac{1}{50}(3,4,5)\right\| = \frac{\sqrt{50}}{50} \neq 1$.
   (b) $-\mathbf{U}$ if $\mathbf{U}$ is: **Yes**, negation preserves magnitude.
   (c) $(-u_1, u_2, -u_3, u_4, -u_5, u_6)$ if $\mathbf{U}$ is: **Yes**, sign changes don't affect norm.
   (d) $\frac{1}{3}(1,-\sqrt{2},\sqrt{3},-\sqrt{3})$: **Yes**, $\left\|\frac{1}{3}(1,-\sqrt{2},\sqrt{3},-\sqrt{3})\right\| = 1$.
2. Determine orthogonality:
   (a) $(1,1,1,1,1)$ and $(-1,-1,-1,-1,-1)$: **No**, dot product -5.
   (b) $(1,1,1,1)$ and $(-1,-1,-1,3)$: **Yes**, dot product 0.
   (c) $(1,1,1)$ and $(-1,2,-1)$: **Yes**, dot product 0.
3. Plane equation: $1x + 0y + 0z = 0$ or $\boxed{x = 0}$.
4. Points on $20x_1 + 10x_2 - 50x_3 = 0$:
   (a) $(0,0,0)$: **Yes**.
   (b) $(0,5,1)$: **Yes**.
   (c) $(-1,2,0)$: **Yes**.
   (d) $c\mathbf{X}$ if $\mathbf{X}$ is: **Yes**.
   (e) $\mathbf{X} + \mathbf{Y}$ if $\mathbf{X}$ and $\mathbf{Y}$ are: **Yes**.
5. Points on $2x_1 + x_2 - 5x_3 = 100$:
   (a) $(0,0,0)$: **No**.
   (b) $(50,0,0)$: **Yes**.
   (c) $(0,100,0)$: **Yes**.
   (d) $(0,100,0) + \mathbf{Y}$ if $2y_1 + y_2 - 5y_3 = 0$: **Yes**.
   (e) $\mathbf{X} + \mathbf{Y}$ if $\mathbf{X}$ and $\mathbf{Y}$ are: **No**.

Part II. Group Problems

1. (a) $(c, c, c, \cdots, c)$
   (b) $nc^2 = 1$ gives $c = n^{-1/2}$.
   (c) $c = n^{-1/2}$ tends to zero as n tends to infinity.